CYBERSECURITY RISK DETECTOR

AI-Powered Network Traffic & Security Analytics Report

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1. Executive Summary

This automated security analytics report details machine learning evaluation of dataset **sample_labeled_dataset.csv**. A total of **600** records were evaluated. The detection pipeline identified **60** statistical anomalies (10.0% of evaluated traffic). Identified records were triaged by risk severity, revealing **2 Critical**, **37 High**, **69 Medium**, and **492 Low** severity indicators.

Metric	Total Records	Critical Risk	High Risk	Medium Risk	Low Risk
Count	600	2	37	69	492

2. Detection Methodology & Engine Details

The analytics engine utilizes an ensemble architecture incorporating unsupervised Isolation Forest for unknown anomaly detection, alongside optional supervised Random Forest classification when validated ground-truth labels are present. Feature vectors undergo schema normalization, robust scaling, and categorical encoding. A rule-assisted risk scoring engine correlates ML anomaly scores with telemetry heuristics including targeted ports (e.g., ports 4444, 3389, 445, 22), data egress volume anomalies, and elevated connection failure ratios.

Important Analytical Disclaimer: *Detected anomalies represent statistical divergence from baseline behavior. Anomalies require human security analyst investigation and do not represent confirmed attacks.*

3. Priority Security Alerts (Top Findings)

Severity	Risk	Source IP	Dest IP : Port	Reason & Indicators
Critical	94/100	192.168.1.204	151.101.1.140 : 8080	Flagged as statistical anomaly by Isolation Forest (score: 94/100). | Destination port 8080 matches
Critical	94/100	192.168.1.204	151.101.1.140 : 8080	Flagged as statistical anomaly by Isolation Forest (score: 94/100). | Destination port 8080 matches
Critical	91/100	192.168.1.206	45.33.32.156 : 8080	Flagged as statistical anomaly by Isolation Forest (score: 91/100). | Destination port 8080 matches
Critical	91/100	192.168.1.206	45.33.32.156 : 8080	Flagged as statistical anomaly by Isolation Forest (score: 91/100). | Destination port 8080 matches
High	77/100	192.168.1.197	185.199.108.153 : 8443	Flagged as statistical anomaly by Isolation Forest (score: 100/100). | Substantial data volume (24.2

High	77/100	192.168.1.132	104.244.42.1 : 443	Flagged as statistical anomaly by Isolation Forest (score: 99/100). | Substantial data volume (27.22
High	77/100	192.168.1.107	185.199.108.153 : 31337	Flagged as statistical anomaly by Isolation Forest (score: 88/100). | Destination port 31337 matches
High	77/100	192.168.1.197	185.199.108.153 : 8443	Flagged as statistical anomaly by Isolation Forest (score: 100/100). | Substantial data volume (24.2
High	77/100	192.168.1.132	104.244.42.1 : 443	Flagged as statistical anomaly by Isolation Forest (score: 99/100). | Substantial data volume (27.22
High	77/100	192.168.1.107	185.199.108.153 : 31337	Flagged as statistical anomaly by Isolation Forest (score: 88/100). | Destination port 31337 matches

4. Recommended Defensive Actions & Mitigation

1. **Isolate & Triage Endpoints:** Cross-reference flagged source IPs with DHCP logs and host asset registries.

2. **Port & Protocol Verification:** Review firewall access control lists for unusual traffic targeting non-standard administrative ports (e.g., 4444, 1337, 3389).

3. **Credential Audit:** For hosts exhibiting high failed-connection ratios, inspect Active Directory / Kerberos / SSH authentication logs for brute-force patterns.

4. **Egress Inspection:** Review NetFlow and Proxy egress logs for unexpected external transfers from hosts flagged with high transfer volumes.

5. **Threat Hunting:** Ingest indicators of compromise (IoCs) into SIEM/EDR for real-time rule correlation across enterprise fleet.

5. Limitations of Analysis

• Machine learning models operate on observed statistical distributions; novel attack techniques or encrypted payload contents may not present visible deviations in metadata features.
• Baseline model drift may occur if network topologies or standard user behaviors shift significantly over time.
• Absence of anomalies does not guarantee the complete absence of stealthy adversary persistence.